EART11300 practical: Minerals in hand specimen - physical properties and mineral identification - answers

By the end of the practical you should be able to:

- Describe a crystal's habit, cleavage and colour
- Determine the relative hardness of different minerals

The PDFs of the 4 videos on Blackboard contain useful background information for this practical. Video 1 explains what a mineral is, video 2 is on crystal habit, video 3 on cleavage in minerals, and video 4 on colour and hardness of minerals.

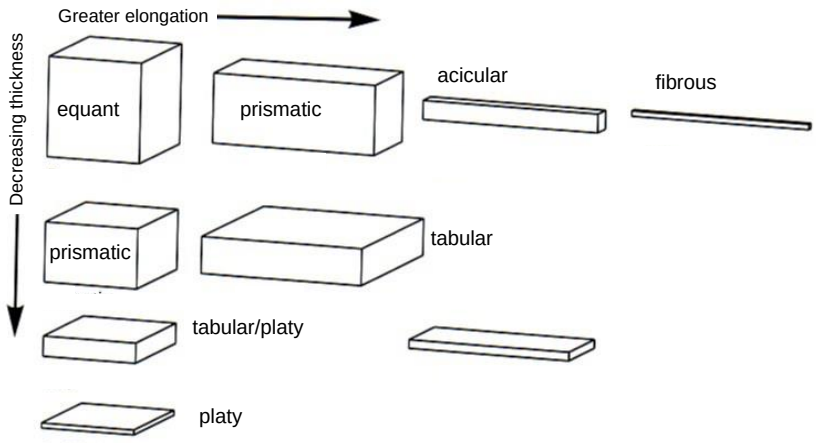
At the start of the practical you will be given a set of 5 minerals, which are yours to keep. Try not to lose them, as you may need them next semester!

Crystal habit

The combination and relative development of crystal faces is called the crystal's 'habit'. It depends partly on crystal structure, but also on the way the crystal has grown, allowing some faces to develop preferentially. Terms used: equant, tabular, platy, prismatic (moderately elongate), acicular (needle-shaped), fibrous. Massive = compact mass of interlocking crystals with no clear division between individual grains.

The mineral samples that you have been given are all single crystals (occasionally there may be 2 or 3 interlocking crystals), but they are not **whole** crystals, as they have been obtained by smashing up larger crystals with a hammer, so unfortunately you are unlikely to see any crystal faces. But if the mineral contains **cleavage**, it is likely to break along cleavage planes when it is hit by the hammer (as these are planes of weakness in the crystal structure). So you will find that some or all of the surfaces of these mineral samples are cleavage planes. You can identify them as cleavage planes as they are flat surfaces that reflect the light well. Four of your samples have cleavage, though one isn't as clear as the others. The other mineral has no cleavage and therefore breaks into irregular fragments like pieces of glass when hit with a hammer. In the following exercise, only look at the 4 minerals whose faces are cleavage planes, and treat them as if they are crystal faces. You will also look at some mineral specimens from the departmental collection which do have nice crystal faces.

Use the figure below to assign a habit term to your 4 minerals with cleavage, and the minerals on display: fluorite, calcite, garnet, quartz, pyrite (M5) (galena, M8, might be more tricky, as most samples contain several interlocking crystals).



Mineral	Habit
biotite (one of your 5)	platy
muscovite (one of your 5)	platy
calcite (one of your 5)	tabular or prismatic (it will vary depending on the precise shape of your sample)
feldspar (one of your 5)	probably too irregular to say
fluorite	equant
calcite	variable – most are closest to prismatic, but could be tabular or equant
garnet	equant
quartz	prismatic
pyrite (M5)	equant
galena (M8)	equant

Cleavage

Many minerals tend to break smoothly along particular planes. This is related to the internal atomic structure: it is a manifestation of weaker bond strength across these planes. Cleavage planes are always parallel to possible (or actual) crystal faces. Like crystal faces, they are related to the crystal's internal symmetry.

Cleavage is more useful as a diagnostic property than crystal shape, since it is not affected by the external conditions which determine crystal shape (the only limitation here being the size of the grains). It is an extremely valuable diagnostic property in thin sections of minerals viewed in the optical microscope.

Cleavage may be developed in 1, 2, 3, 4 or 6 directions. Cleavage may be classified qualitatively as perfect, good, moderate or poor, and also in terms of the angular relationship between cleavage planes. In the absence of cleavage, crystal breakage results in the development of fracture, which may be irregular, or in some cases, conchoidal (producing characteristically curved surfaces, as with broken glass).

Use the table and figure below to determine the type of cleavage that each of the minerals has. This is not easy for some of the display minerals, since most of the surfaces you can see are crystal faces, so you will need to look for planar cracks inside the crystals. Note also that the quality of cleavage (how 'good' it is) varies from mineral to mineral. If cleavage is 'perfect', the mineral will easily split along the cleavage planes, but if it is 'poor', then it won't split easily. In between 'perfect' and 'poor' are 'good' and 'moderate'. These terms are only qualitative, so don't worry if you are not sure whether to say 'good' or 'perfect'.

Put your 4 minerals with cleavage in order of how good the cleavage is.
biotite, muscovite – best
calcite
feldspar

Two of your 5 minerals are the same as two of the display minerals. Which ones? quartz and calcite

Table showing the numbers and angles of cleavages possible in crystals

Number of cleavage directions	Description
0 quartz, garnet, pyrite	No cleavage Mineral breaks into irregular masses with no flat surfaces
1 biotite, muscovite	Basal cleavage Mineral splits apart along flat sheets

2 at 90° Feldspar (there is also another, poor, cleavage at 90° to these, that you may be able to see)	Prismatic cleavage Elongate with rectangular cross sections
2 not at 90°	Prismatic cleavage Elongate with parallelogram cross sections
3 at 90° galena	Cubic cleavage Mineral breaks into cubes
3 not at 90° calcite	Rhombohedral cleavage Mineral breaks into rhombohedra
4 fluorite	Octahedral cleavage Mineral breaks into octahedra
6	Dodecahedral cleavage Mineral breaks into dodecahedra

Note that the above terms refer to equivalent cleavage planes.

More details and images showing cleavage in your 5 minerals:

Calcite:
See https://sketchfab.com/3d-models/calcite-f0ab56495a5d46478f01090dbc6e88f4. This shows a cleavage fragment of calcite. The faces of this fragment are cleavage planes. There are 6 faces, but the top is in the same orientation as the bottom, so the top and bottom count as 1 cleavage direction; the front and back are in the same orientation and so these count as 1 direction; and the 2 sides are also in the same orientation and so count as 1 direction. So overall there are 3 cleavage directions. Since the shape of each face is a parallelogram, not a rectangle, the angles are not at 90°. If you look carefully at the faces, you will see evidence of the cleavage cracks inside the crystal. This is where the crystal would split if you hit it with a hammer, and it would become a smaller version of itself. Here's another example: https://sketchfab.com/3d-models/calcite-d8117fb0d4b44dbe8218d81887484039. Calcite is generally defined as having "perfect" cleavage. Both these examples have a prismatic habit.

Feldspar:
See https://sketchfab.com/3d-models/orthoclase-feldspar-a8ab6e5cc9cd4378aaef339e4e5684a6. This might look like it has been cut into a square prism. But in fact you are looking at the actual crystal faces. The cleavage in feldspar is in the same orientation as these faces, i.e. there are 2 cleavage directions at 90° to each other. Feldspar has "good" cleavage. This particular crystal has a prismatic habit.

Mica:
See https://sketchfab.com/3d-models/muscovite-4b3f57509ddb4907b4b39da578af2c94 for an example. This example is muscovite, which should be a silvery colour. It looks dark in this photo because it is quite thick. If you look at the top of the image you will see its characteristic silvery colour. As you move the image about, you will see the light reflecting off it like a mirror, due to its perfect cleavage. This occurs in only one direction, meaning mica easily splits into flakes, and has a platy habit. https://sketchfab.com/3d-models/muscovite-04-20-2020-8588fb59d0f2492cb8dc3fd7e423a88f shows a more characteristic colour, but doesn't show the shiny cleavage surfaces so well.
https://sketchfab.com/3d-models/biotite-3094-07-06-2020-1e06947d652846ec9e65da8faadde47a shows the black colour of biotite, but again doesn't look very shiny in this image. All these micas have a platy habit.

Quartz:
Quartz does not have any cleavage, so if you break up a large crystal, it will not split along flat planes, and instead breaks into irregular pieces, such as https://sketchfab.com/3d-models/quartz-40dbf45ad43b485fa9188a5507e8fed4.

Colour

The impression of colour in minerals results from the preferential absorption of certain wavelengths from white light, so that the complementary colour is seen. Although colour is the mineral property which makes the most immediate impact, its causes are complex, and it must be used as a diagnostic property with some care.
Some minerals are intrinsically coloured because of their chemical composition (this applies particularly to minerals containing essential Ti, V, Cr, Mn, Fe, Ni, Co and Cu) and they produce a characteristic colour (streak) when powdered. In practice this is generally achieved by drawing the mineral across an unglazed porcelain plate (only works if the mineral is softer than the plate,
H < 6 ½).
Most other minerals are essentially colourless in powdered form (i.e. produce a white streak), but may occur as distinctly coloured crystals in the natural state due to impurities. Such minerals may be very variable in body colour depending on the amount and type of impurity in particular occurrences (e.g. fluorite, quartz, etc.). Colour may also be produced by defects in the crystal lattice. Some minerals develop an iridescent film or tarnish as the result of surface oxidation (e.g. chalcopyrite).
Others produce a 'play of colours' or schiller effect due to internal reflection from minute platy inclusions or other discontinuities such as twinning. This is seen in some varieties of feldspar, and also in opal (which consists of minute spheres of amorphous SiO_2), when it is termed opalescence.

You can now finish identifying your minerals:

You will have found that 2 of your minerals have the same habit and cleavage, but have very different colours. One is black and the other is silvery. If you peeled away a thin layer of the black one it would either still look black, or be brown and transparent, depending on how thin a layer you can peel (try it…), but if you do the same with the silvery one it will be colourless (and therefore transparent). The black one is black because it contains **iron (Fe^{2+})**; the colourless one does not contain iron. These are both minerals in the **mica** group. **Biotite** has the composition **$K(Mg,Fe)_3[AlSi_3]O_{10}(OH)_2$** and **muscovite** has the composition **$KAl_2[AlSi_3]O_{10}(OH)_2$**. The amount of Fe in biotite is variable; the more Fe, the darker the colour. Muscovite can contain impurities which give it a hint of colour.

The other minerals are all transparent or translucent (but small pieces would be transparent) in shades of white, pale grey and orangey-pink (powdered samples would be colourless). The mineral without cleavage is **quartz**, which has the composition **SiO_2**. Pure quartz is completely colourless and transparent, like glass. Small amounts of impurities are possible in quartz, but not common. "Milky" quartz is common, the white colour caused by tiny fluid inclusions.

Calcite $CaCO_3$ is also colourless when pure, and white when it contains fluid inclusions. Any pink colouration in your crystal is from iron: Fe^{2+} can substitute for Ca^{2+} in the crystal structure, and may later oxidise to Fe^{3+} to give a rusty colour.

Feldspar (variable composition; the end-member closest to your sample is **orthoclase $KAlSi_3O_8$**) has a pinkish colour, due to iron impurities. You'll see that the colour is not uniform, and looks more pink adjacent to little fractures running through the crystal. This indicates variable amounts of oxidation of the Fe^{2+} impurity to Fe^{3+}.

Observe the beautiful play of colours (known as schiller) shown by the large polished crystal of labradorite feldspar on display. This is due to the presence of a submicroscopic periodic lamellar structure in the crystal that causes light to diffract. The lamellae are produced by the process of exsolution that takes place as the mineral cools from high temperature.

Examine the variation in colour that is shown by the quartz crystals on display. These colours are caused by the incorporation of very small amounts of impurities, e.g. Fe in citrine (yellow), Ti, Fe, Mn in rose quartz. Sometimes the impurity is associated with a missing electron, to maintain charge balance, e.g. $Si^{4+} + e^- \rightarrow Fe^{3+}$ in amethyst, $Si^{4+} + e^- \rightarrow Al^{3+}$ in smoky quartz. These missing electrons are called colour centres.

Hardness

Defined as resistance to scratching. Hardness depends on bond density, and can vary with direction in a mineral because there are different numbers and strengths of bonds in different directions. It is usually measured by reference to a set of standard minerals – Mohs' Scale of Hardness (1812). This is a qualitative test: each mineral is capable of scratching the minerals with lower hardness.

1. talc
2. gypsum
3. calcite
4. fluorite
5. apatite
6. orthoclase
7. quartz
8. topaz
9. corundum
10. diamond

Using the hardness set (and Mohs' scale), confirm the hardnesses of the implements provided (including your finger nail!). You can then use these implements to determine/check the hardness of your 5 minerals, and the set of unknown minerals in the next question (but not the nice specimens that you have been using as examples of habit, cleavage and colour!). Note that if you try to scratch a harder mineral with a softer one, the softer one can leave a powder trail on the harder one which makes it look like it scratched the harder one.

Hardnesses are:
fingernail 2 – 2½; copper coin 3½ - 4 ; steel knife 5+ ; window glass 5½; steel scalpel 6½
Mica is harder than a fingernail, at approx. 2.5 - 3

Why do you think that your glassy-looking quartz crystal is quite a lot harder than actual glass?
Quartz is harder because the atoms in its regular crystal structure are more closely packed than in the irregular structure of glass.

Identification of unknown minerals

Using the table of physical properties provided, identify minerals A – M. Note that there is no mineral I, and there may be more than one sample of some minerals.

Unknown minerals

A. galena
B. barite
C. gypsum
D. pyrite
E. calcite
F. quartz
G. hematite
H. gypsum
I. -
J. barite
K. halite
L. quartz
M. fluorite